

Trivia Quiz System - Final Project Documentation

This project is a Trivia Quiz System developed in C++ using Object-Oriented Programming principles. The system contains two applications: a player application and an admin application. The player can answer quiz questions, save progress, and view the leaderboard. The admin can manage questions and leaderboard data.

1. Team Tasks

Student 1	Quiz gameplay, leaderboard system, file handling
Student 2	Admin system, progress saving, question management

2. Data Structures and Classes

Question

- Stores quiz question text, options, correct answer, and difficulty.

Player

- Stores player name and score.

GamePlayer : public Player

- Extends Player class and adds save/load progress functionality.

Admin

- Manages quiz questions and leaderboard information.

3. Relationships Between Classes

The project uses inheritance relationships:

- GamePlayer inherits from Player
- Admin inherits from User

Inheritance was used to reuse code and extend functionality.

4. File Structure

questions.txt

Stores all quiz questions.

leaderboard.txt

Stores player scores.

progress.txt

Stores saved game progress.

5. Application Functionalities

Main Application:

- Start quiz
- Continue saved progress
- View leaderboard
- View questions

Admin Application:

- Add questions
- Delete questions
- View leaderboard
- Reset progress

6. OOP Concepts Used

- Classes and Objects
- Inheritance
- Encapsulation
- STL Vectors
- Templates
- Operator Overloading
- File Handling

7. Conclusion

This project demonstrates the implementation of a complete C++ application using Object-Oriented Programming concepts. The project includes inheritance, templates, STL containers, operator overloading, and file handling.